Hawai‘i – Inland flooding ESF-6 playbook

Data as of 2026-10-27

County: Hawai‘i · **Hazard:** Inland flooding

Planning aid. Supports, does not replace, county emergency plans. Not an official shelter list.

1. Summary

People at risk **65,518**	Shelter demand (central) **3,276**	Shelter demand low – high **1,310 – 7,207**
Limited English (C) **1,459**	Power-dependent (M) **484**	Disability (I) **10,217**
Age 65+ (I) **14,992**	People without a vehicle (T) **4,025**	Known 2026 sites usable **10**

Shelter-seeking rates: low 2%, central 5%, high 11.0%. CMIST counts are people at risk, from census-tract estimates; one person can appear in more than one count.

2. Scope: county total vs. one community

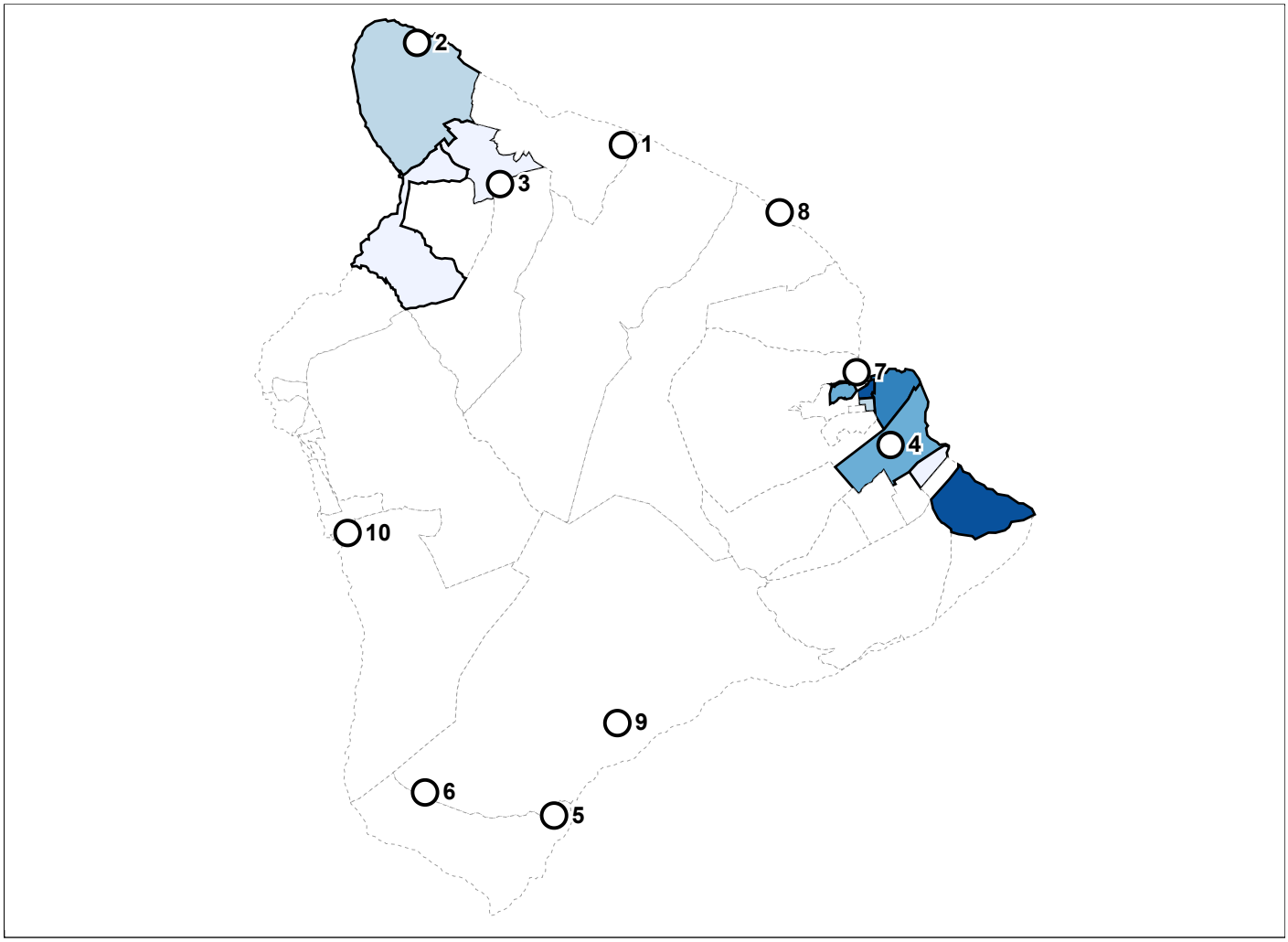
County total = upper bound. It assumes every at-risk tract is hit at once (top 25% of each county's tracts by NRI score). Real events usually hit fewer communities.

Largest single community (highest-demand census tracts)

Tract	At risk	Shelter low / central / high	Limited English	Power-dep.	Disability	65+	No vehicle
15001021014	8,135	163 / **407** / 895	260	49	834	1,504	0
15001020802	7,010	140 / **350** / 771	50	64	1,087	1,974	358
15001020600	6,802	136 / **340** / 748	67	45	1,261	1,394	387

Plan a realistic one-community event with the first row.

3. Map: AFN need and known sites



- AFN need 0–1,143 people (class 1 of 5)
- AFN need 1,143–1,430 people (class 2 of 5)
- AFN need 1,430–1,558 people (class 3 of 5)
- AFN need 1,558–1,760 people (class 4 of 5)
- AFN need 1,760–2,219 people (class 5 of 5)
- Solid outline: tract in hazard footprint (at risk)
- Dashed outline: not in footprint
- Usable
- Check: flood (SFHA)
- Check: flood (tract-level)
- Flood risk undetermined (zone D)
- Excluded: tsunami zone

AFN need = power-dependent + disability + limited English + no-vehicle people at risk (people may be counted more than once). Numbers on markers match the sites table. No basemap: tract outlines only.

4. Known sites (opened in 2026 events)

#	Status	Site	Address	Reason	Times opened 2026
1	✓ **Usable**	Honokaʻa Gym	45-541 Lehua St., Honokaʻa	FEMA zone X: outside Special Flood Hazard Area	1
2	✓ **Usable**	Ikuo Hisaoka Gym	54-382 Kamehameha Park Road, Kapaʻau	FEMA zone X: outside Special Flood Hazard Area	1
3	✓ **Usable**	Kahilu Town Hall	67-1182 Lindsey Road, Waimea	FEMA zone X: outside Special Flood Hazard Area	1
4	✓ **Usable**	Keaʻau Armory	16-512 Volcano Road, Keaʻau	FEMA zone X: outside Special Flood Hazard Area	1
5	✓ **Usable**	Nāʻālehu Elementary School Gym	95-5545 Mamalahoa Highway, Nāʻālehu	FEMA zone X: outside Special Flood Hazard Area	1
6	✓ **Usable**	Ocean View Community Center	92-8924 Leilani Circle, Ocean View	FEMA zone X: outside Special Flood Hazard Area	1
7	✓ **Usable**	Puʻuʻeo Community Center	145 Wainaku St., Hilo	FEMA zone X: outside Special Flood Hazard Area	1
8	✓ **Usable**	Pāpaʻaloa Park Annex	35-1994 Government Main Road, Pāpaʻaloa	FEMA zone X: outside Special Flood Hazard Area	1
9	✓ **Usable**	Robert N. Herkes Gym	96-1219 Kamani St., Pāhala	FEMA zone X: outside Special Flood Hazard Area	1
10	✓ **Usable**	Yano Hall	82-6156 Māmalahoa Highway, Captain Cook	FEMA zone X: outside Special Flood Hazard Area	1

Candidate public schools in this county usable for this hazard: **54**. Sites are places a county has opened before; not every site opens in every event. Always follow current county announcements.

5. Needs and proposed partners

Central estimate. Roles are **proposed – to be confirmed with partner**. Pledged capacity stays blank until a partner confirms it.

Need	Quantity	Unit	Proposed lead	Proposed support	Pledged	Gap
Cots / sleeping space	3,276	cots	county_em	dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies	—	confirm with partner
Floor space (≤72 h)	65,518	sq ft	county_em	dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies	—	confirm with partner
Floor space (≤2 weeks)	131,036	sq ft	county_em	dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies	—	confirm with partner
Water	3,276	gal/day	red_cross	dhs; county_em; doe; salvation_army; hawaii_voad; faith_based; business_community	—	confirm with partner
Meals	6,552	meals/day	red_cross	dhs; county_em; doe; salvation_army; hawaii_voad; faith_based; business_community	—	confirm with partner
Toilets (≤72 h)	82	toilets	county_em	dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies	—	confirm with partner
Toilets (≤2 weeks)	164	toilets	county_em	dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies	—	confirm with partner
Shelter staff	223	staff per shift	county_em	dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies; hawaii_voad; oha; salvation_army; philanthropy; business_community	—	confirm with partner
Power outlets for power-dependent residents	25	outlets	doh	county_em; health_care; utilities	—	confirm with partner
Accessible transport	4,025	people in no-vehicle households at risk	county_em	dhs; hiema; dot; business_community; other_agencies	—	confirm with partner
Pet space – dog households	522	households with a dog	county_em	hawaii_voad	—	confirm with partner
Pet space – cat households	304	households with a cat	county_em	hawaii_voad	—	confirm with partner
Children's area	703	children in shelter	county_em	dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies; doh; health_care; philanthropy	—	confirm with partner
Interpreters – Ilocano	19	people with limited English in shelter	dhs	county_em; hiema; hspls; doh; oha; faith_based; other_agencies	—	confirm with partner
Interpreters – Other English-based Creole languages	11	people with limited English in shelter	dhs	county_em; hiema; hspls; doh; oha; faith_based; other_agencies	—	confirm with partner
Interpreters – Japanese	8	people with limited English in shelter	dhs	county_em; hiema; hspls; doh; oha; faith_based; other_agencies	—	confirm with partner
Interpreters – Spanish	7	people with limited English in shelter	dhs	county_em; hiema; hspls; doh; oha; faith_based; other_agencies	—	confirm with partner
Interpreters – Tagalog, Filipino	5	people with limited English in shelter	dhs	county_em; hiema; hspls; doh; oha; faith_based; other_agencies	—	confirm with partner
Interpreters – Chinese	4	people with limited English in shelter	dhs	county_em; hiema; hspls; doh; oha; faith_based; other_agencies	—	confirm with partner
Interpreters – Hawaiian	3	people with limited English in shelter	dhs	county_em; hiema; hspls; doh; oha; faith_based; other_agencies	—	confirm with partner
Interpreters – Marshallese	3	people with limited English in shelter	dhs	county_em; hiema; hspls; doh; oha; faith_based; other_agencies	—	confirm with partner

6. Decision points

Proposed – supports, does not replace, county plans.

When	Trigger	Action	Owner (proposed)	Support
BEFORE	Annual pre-positioning review (no official product) County (sets the date)	Mark shelters inside a FEMA Special Flood Hazard Area or zone D for site check; choose alternates.	County Emergency Management (each county)	doe, dhs
DURING	Flash Flood Warning NOAA NWS Honolulu	Open only shelters outside the SFHA; accessible transport where roads allow; multilingual messages.	County Emergency Management (each county)	dot, red_cross, dhs
AFTER	Warning cancelled or all clear County / NWS / PTWC	Recovery sheltering and feeding for displaced households.	American Red Cross	hawaii_voad, salvation_army, faith_based

7. Priority outreach and languages

Top 10 census tracts by AFN need (tract-level counts only)

#	Tract	At risk	Power-dep.	Disability	Limited English	No vehicle	AFN need
1	15001020500	5,552	31	836	149	1,203	2,219
2	15001021108	6,155	43	1,497	56	486	2,082
3	15001020600	6,802	45	1,261	67	387	1,760
4	15001020300	4,167	32	670	31	991	1,725
5	15001020802	7,010	64	1,087	50	358	1,558
6	15001021013	5,059	41	955	207	286	1,488
7	15001021800	6,435	60	1,134	182	55	1,430
8	15001020701	5,246	44	918	261	98	1,320
9	15001021014	8,135	49	834	260	0	1,143
10	15001021705	6,507	41	575	94	55	765

Languages: people who speak English less than "very well" (Hawaii)

Language	People 5+	90% MOE
Ilocano	3,409	±869
Other English-based Creole languages	2,040	±740
Japanese	1,408	±504
Spanish	1,217	±499
Tagalog, Filipino	924	±368
Chinese	651	±326
Hawaiian	637	±264
Marshallese	538	±303

Source: ACS PUMS 2019-2023 5-year (validated against DBEDT Data Book 2024, Table 1.44). Pre-approved message templates: none yet; no machine translation without human review.

8. Sources and assumptions

Planning factors

Factor	Value	Source or team assumption
Shelter-seeking rate – Hurricane	0.02 / 0.03 / 0.05	South Carolina EMD, 2026 Hurricane Response Annex G (planning factors 2%, 3%, 5%)
Shelter-seeking rate – Wildfire	0.02 / 0.05 / 0.112	high = observed Maui 2023: 1,418 people in shelters on Aug 11, 2023 (Maui Now) / Lahaina CDP population 12,702 (2020 Census) = 0.1116, rounded to 3 decimals = 0.112 (so the calibration event falls inside the band). low and central are team_assumption.
Shelter-seeking rate – Tsunami	0.02 / 0.05 / 0.11	team_assumption: central 5%; low 2% = hurricane planning factor (SCEMD); high 11% = observed Maui 2023 no-notice total-loss rate
Shelter-seeking rate – Inland flooding	0.02 / 0.05 / 0.11	team_assumption: central 5%; low 2% = hurricane planning factor (SCEMD); high 11% = observed Maui 2023 no-notice total-loss rate
Shelter-seeking rate – Coastal flooding	0.02 / 0.05 / 0.11	team_assumption: central 5%; low 2% = hurricane planning factor (SCEMD); high 11% = observed Maui 2023 no-notice total-loss rate
Footprint – Hurricane	all	Hurricanes affect whole islands (Lala, Lowell 2026).
Footprint – Tsunami	tsunami_zone_share	Population share inside the State standard tsunami evacuation zone. Extreme zone is reported separately.
Footprint – Wildfire	nri_top_share_by_county (25%)	Counties lead mass care, so each county plans for its own highest-risk tracts (team_assumption).
Footprint – Inland flooding	nri_top_share_by_county (25%)	Counties lead mass care, so each county plans for its own highest-risk tracts (team_assumption).
Footprint – Coastal flooding	nri_top_share_by_county (25%)	Counties lead mass care, so each county plans for its own highest-risk tracts (team_assumption).
floor sqft 72h	20	Florida DEM Sheltering Guidance Aid (2010): emergency evacuation shelter, <=72 h
floor sqft 2wk	40	Florida DEM (2010): short-term shelter, <=2 weeks
water gal per day	1	Florida DEM (2010)
meals per day	2	Florida DEM (2010): two hot meals a day plus snacks
persons per toilet short term	20	Florida DEM (2010): short-term shelter, <=2 weeks
power outlets per power dependent person	1	team_assumption
persons per toilet 72h	40	Florida DEM (2010): emergency evacuation shelter, <=72 h
Staffing ratios	client registration 1:165, dormitory management 1:100, meal service 1:110, mental health 1:250, public health medical 1:200, security 1:100, sanitation 1:200, children area 1:12	One staff member per N shelter residents, minimum 1 when the shelter is open. Florida DEM (2010) staffing metrics; the source marks them as recommendations.
dog owning household share	0.446	AVMA U.S. pet ownership statistics (2020 data). Verified in 2022 AVMA Pet Ownership and Demographics Sourcebook, p. 3: 'Dog ownership increased by 6% from 38.4% (2016) to 44.6% (2020)'. The first url did not resolve when checked 2026-10-27.
cat owning household share	0.26	AVMA (2020 data). Verified in 2022 AVMA Pet Ownership and Demographics Sourcebook, p. 3: 'Cat ownership increased slightly from 25% in 2016 to 26% in 2020'.

Data sources

Source	Fetched	URL / file
CDC/ATSDR Social Vulnerability Index 2022, census tracts	2026-10-27	https://onemap.cdc.gov/onemapservices/rest/services/SVI/CDC_ATSDR_Social_Vulnerability_Index_2022_USA/FeatureServer/2

Source	Fetched	URL / file
FEMA National Risk Index v1.20, census tracts – Hurricane	2026-10-27	https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Hurricane_Hazard_Type_Risk_Index_
FEMA National Risk Index v1.20, census tracts – Wildfire	2026-10-27	https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Wildfire_Hazard_Type_Risk_Index_Ra
FEMA National Risk Index v1.20, census tracts – Tsunami	2026-10-27	https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Tsunami_Hazard_Type_Risk_Index_R
FEMA National Risk Index v1.20, census tracts – Inland Flooding	2026-10-27	https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Inland_Flooding_Hazard_Type_Risk_I
FEMA National Risk Index v1.20, census tracts – Coastal Flooding	2026-10-27	https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Coastal_Flooding_Hazard_Type_Risk_
Tsunami Evacuation Zones (Statewide GIS)	2026-10-27	https://geodata.hawaii.gov/arcgis/rest/services/Hazards/MapServer/2
Extreme Tsunami Evacuation Zones (Statewide GIS)	2026-10-27	https://geodata.hawaii.gov/arcgis/rest/services/Hazards/MapServer/12
HHS emPOWER – electricity-dependent DME, county level	2026-10-27	https://services2.arcgis.com/ZQ4jTQn6k7VPXEwO/arcgis/rest/services/HHS_emPOWER_REST_Service_Public/FeatureServer/2
Hawaii Statewide GIS – State DFIRM (FEMA flood hazard areas, 6/15/26)	2026-10-27	https://geodata.hawaii.gov/arcgis/rest/services/Hazards/MapServer/6

Source	Fetched	URL / file
County shelter activations 2026 (official county releases)	see table	data/sources/county_shelter_activations.csv
ACS PUMS 2019-2023 5-year (detailed languages)	2026-10-27	https://www2.census.gov/programs-surveys/acs/data/pums/2023/5-Year/csv_phi.zip
DBEDT State of Hawaii Data Book 2024, Table 1.44 (validation)	2026-10-27	https://files.hawaii.gov/dbedt/economic/databook/2024-individual/01/014424.pdf
Partner list: Hawaii DHS, HAIC 2026 ESF-6 challenge statement	2026-10-15	HAIC_AI Challenge_Submission_DHS_Disaster_2026.pdf
pre season hurricane		https://www.nhc.noaa.gov/cphcmission.php
hurricane watch		https://www.nhc.noaa.gov/aboutgloss.shtml
hurricane warning		https://www.nhc.noaa.gov/aboutgloss.shtml
tsunami warning		https://www.tsunami.gov/?page=message_definitions
flash flood warning		https://www.weather.gov/safety/flood-watch-warning
red flag warning		https://www.weather.gov/safety/wildfire-ww
coastal flood warning		https://www.weather.gov/okx/marinedef

Privacy: aggregate census-tract and county data only. No individual records. Public facility names and addresses only.